TITANIC SURVIVAL PREDICTION USING ARTIFICIAL INTELLIGENCE

Name: Somarouthu Devi Sathwika

Intern ID: CITS1935

Project Title:

Titanic Survival Prediction using Artificial Intelligence

Objective:

The objective of this project is to predict whether a passenger survived the Titanic disaster using Artificial Intelligence and Machine Learning techniques based on features such as age, gender, passenger class, fare, and embarkation point.

Tools and Technologies:

• Python

• Google Colab

• Scikit-learn

• Pandas

• NumPy

Dataset:

The Titanic dataset is used in this project. It contains passenger details such as:

PassengerId

Pclass

Name

Sex

Age

SibSp

Parch

Ticket

Fare

Cabin

Embarked

Survived (Target variable)

Methodology:

Loaded the Titanic dataset.

Performed data exploration and visualization.

Handled missing values in Age, Embarked, and Fare columns.

Encoded categorical variables into numerical format.

Split the dataset into training and testing sets.

Trained a Random Forest Classifier model.

Predicted survival of passengers.

Evaluated model performance using accuracy and classification report.

Output:

- Dataset preview displayed successfully
- Accuracy: 82.12%
- Precision, Recall, and F1-score evaluated for both classes
- Predicted Result: Not Survived

Result:

The Artificial Intelligence model successfully predicted Titanic passenger survival with an accuracy of 82.12%. The model performed well in classifying both survived and not survived passengers using the Random Forest Classifier.

Conclusion:

This project demonstrates how Artificial Intelligence and Machine Learning can be used to analyze historical data and predict survival outcomes. The model provided good accuracy and reliable predictions, showing the effectiveness of classification algorithms in real-world datasets.